

Context-aware child-directed speech detection from long-form recordings

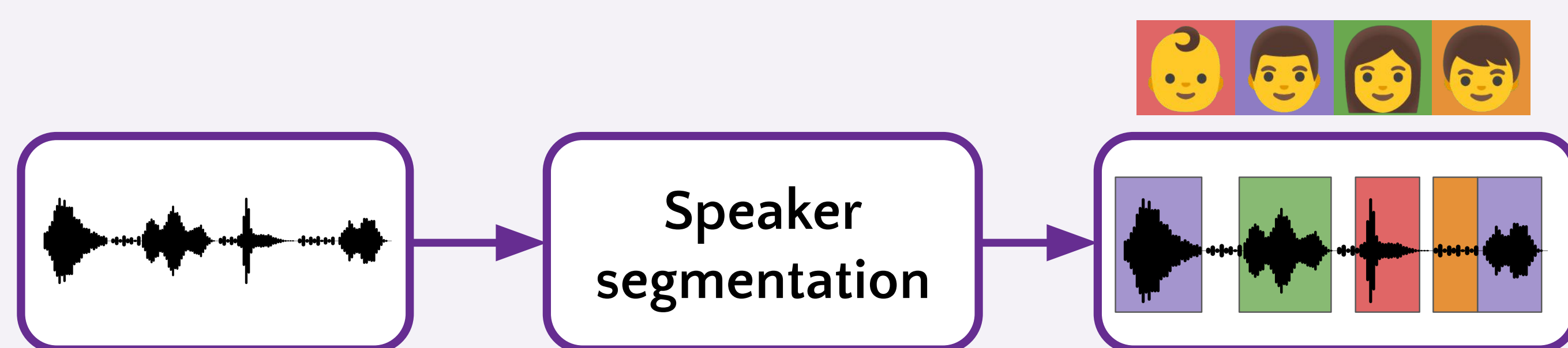
Background: The use of *child-directed speech* (characterized by higher pitch, slower tempo, frequent repetitions, etc.) predicts later language skills. Yet, its role remains debated and automatic tools for detecting it are still unreliable.

Question: Can we build an algorithm that automatically classifies *who's talking to whom* in child-centered recordings?

The pipeline

1. Who's talking when ?

BabyHuBERT (Charlot, et al, 2026)

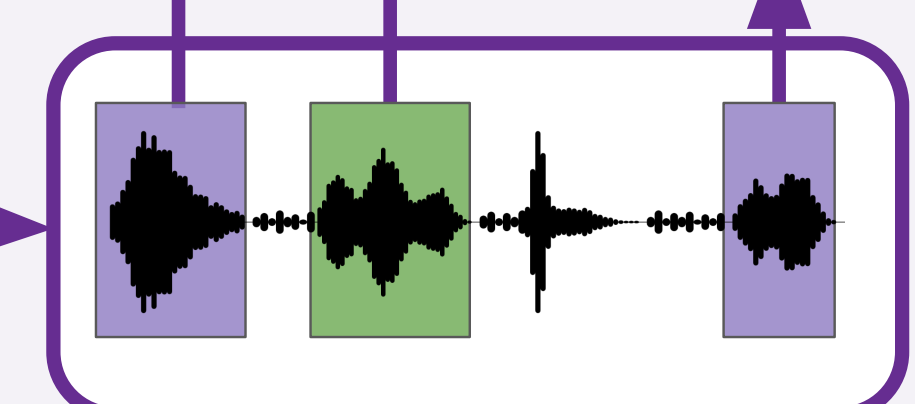


2. Who's talking to whom ?

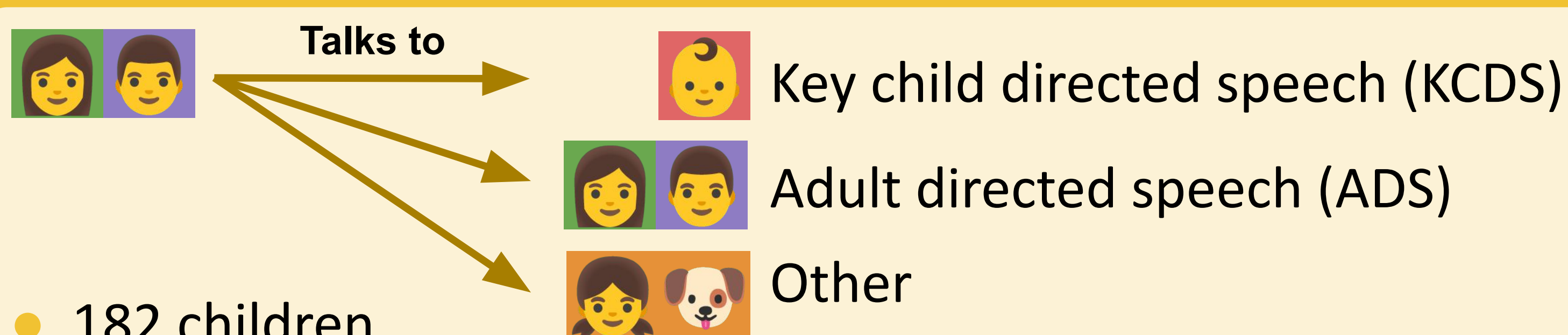
Adult speech

Addressee classification

KCDS: key child-directed speech
ADS: adult-directed speech
Other



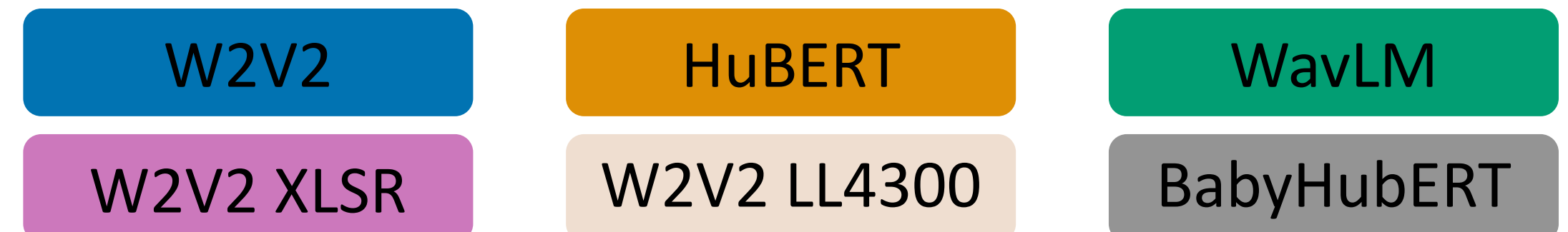
Data



- 182 children
- Train, val, test (19h) : English (56%), French (29%), Yeli Dnye (11%), and Tsimane (3.2%)
- Held out (3h) : Mayan (52%), English (48%)
- Diverse socio-cultural settings

Experiments

Varying encoders (RQ1)



Varying context size (RQ2)

Context duration = 0s:

Context duration > 0s:

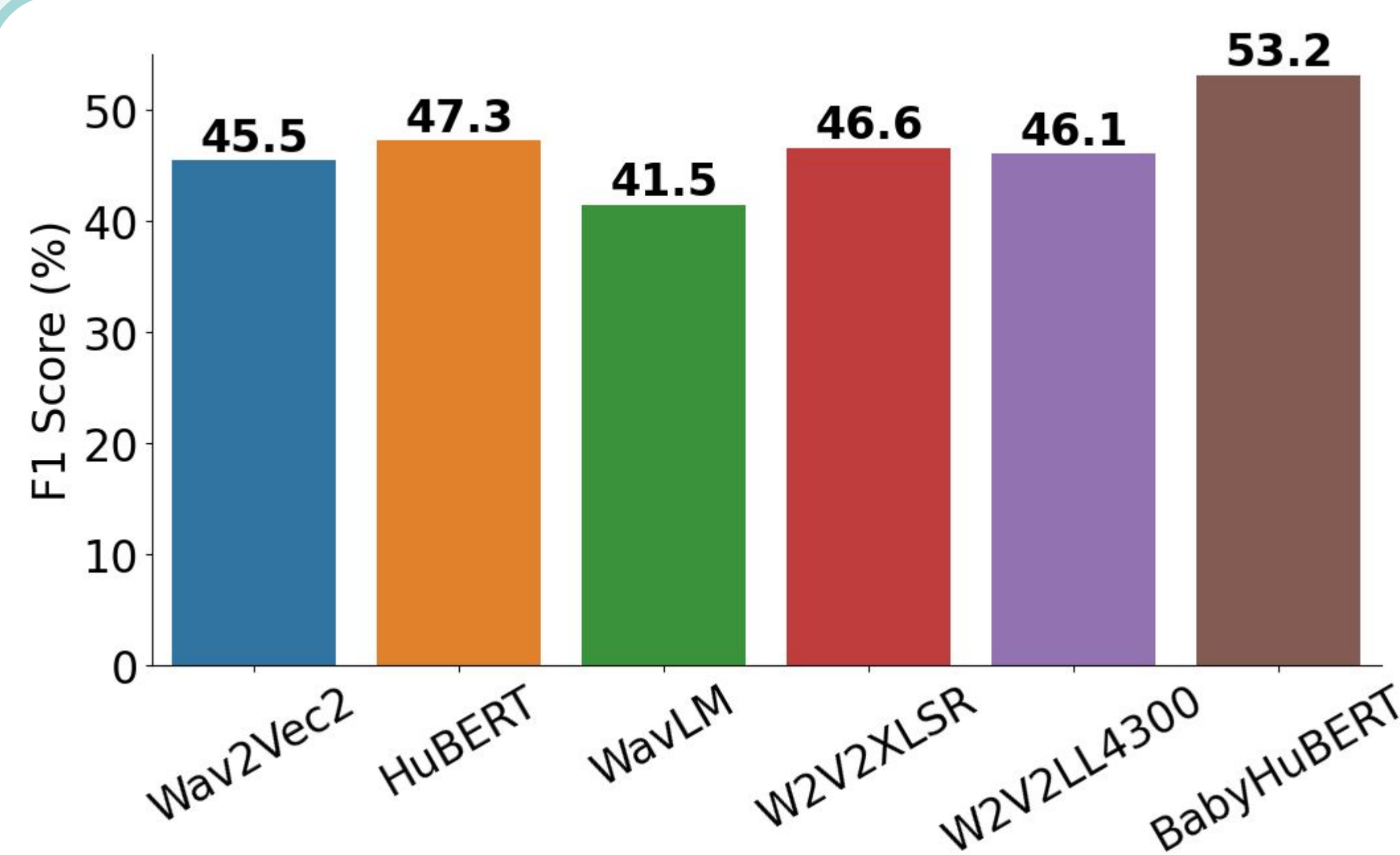
Target segment

Contextual cues

RQ1: Which encoder works best?

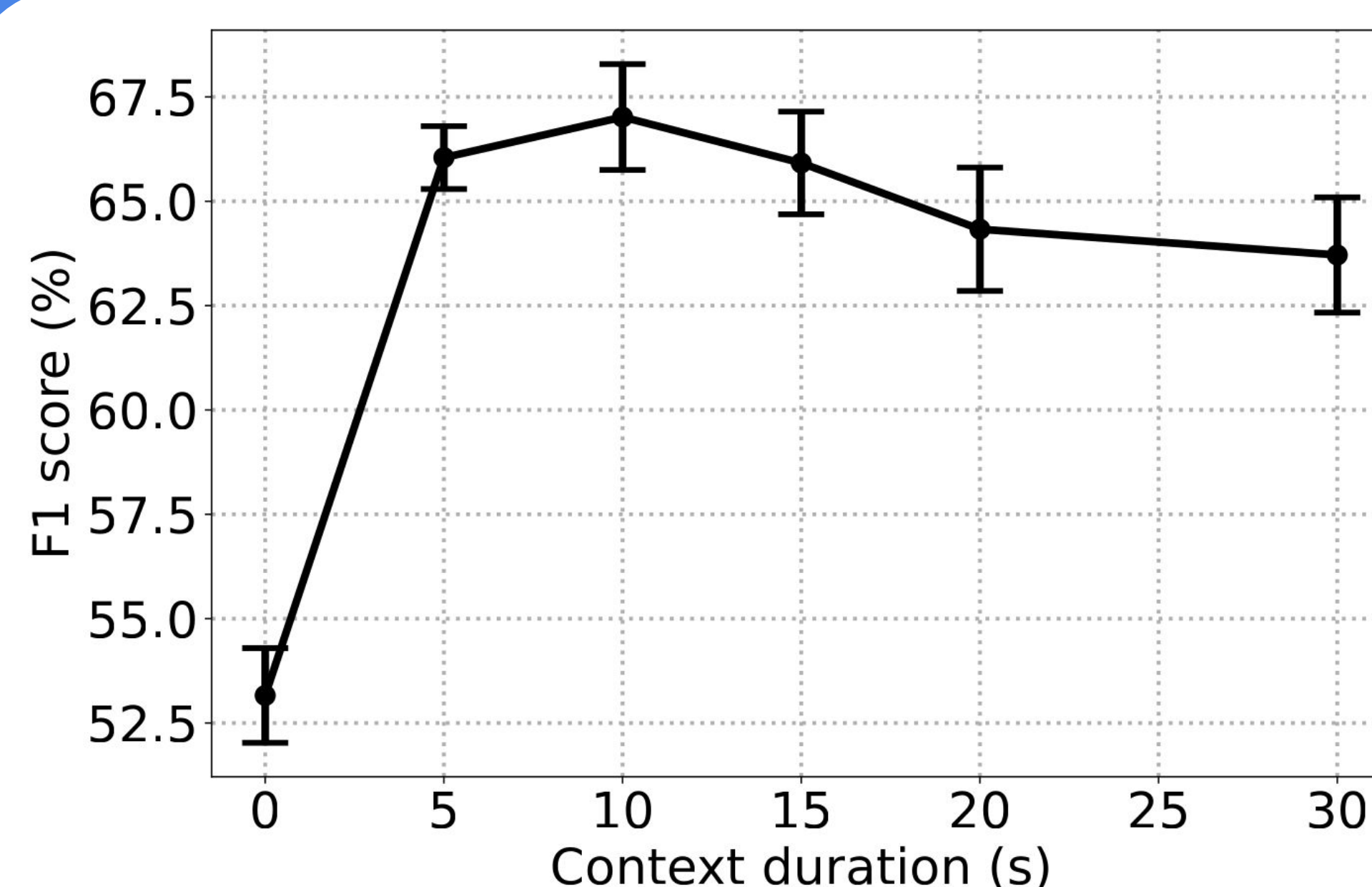
Different pre-training sets:

- English adult speech (W2V2, HuBERT, WavLM)
- Multilingual adult speech (W2V2 XLSR)
- English child + adult speech (W2V2 LL4300)
- Multilingual child + adult speech (BabyHuBERT)



RQ2: How much context helps?

- 13.8% absolute improvement in F1 score by using the context
- Largest gains occurs in the first 5s (+11.8%), but gradually declines >10s (up to -3.3%)



RQ3: Fully automatic pipeline

- Average F1 score:
English : 80.8%
Mayan : 58.3%
- 35.5% drop, propagation of errors from VTC2.0
- BabyHuBERT-Addressee still outperforms rule-based method

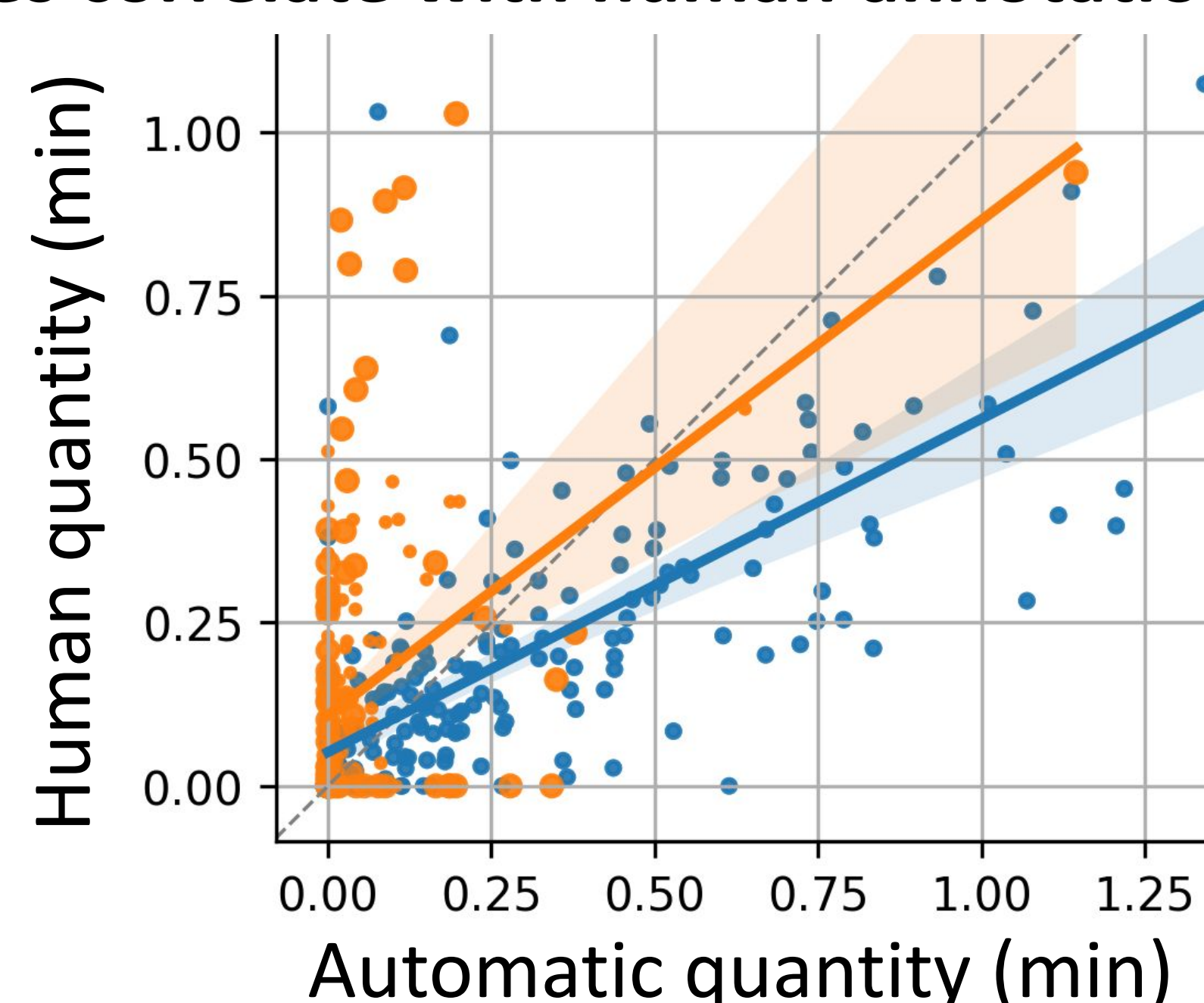
Model	F1-scores (%) ↑		
	KCDS	OHS	Ave.
<i>Human segmentation</i>			
Rule-based	37.9	32.3	35.1
BabyHuBERT-addressee	64.1	84.1	74.1
<i>VTC 2.0 segmentation</i>			
Rule-based	36.3	14.9	25.6
BabyHuBERT-addressee	43.4	33.8	38.6

Bonus RQ4: Do quantity estimates correlate with human annotation?

Quantity of key-child directed speech

Pearson's r correlation

- English $r=0.69$ ($p < 1e-34$)
- Mayan $r=0.35$ ($p < 1e-7$)



Future questions

- How does quantity of child-directed speech vary across socio-cultural settings?
- Do automatic estimates predict later language skills
- What's the effect of KCDS on spoken language modeling?



Paper and code available!

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